SOCIAL NETWORK ANALYSIS ON MEDIA BIAS IN NEWS ARTICLES

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BY

ABHISHEK SINGH
ENROLMENT NO. 20/11/EC/037



SCHOOL OF ENGINEERING
JAWAHARLAL NEHRU UNIVERSITY
NEW DELHI-110067

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JAWAHARLAL NEHRU UNIVERSITY
SCHOOL OF ENGINEERING

DECLARATION

I declare that the project entitled **"Social Network Analysis on Media Bias in News Articles",** submitted by me in fulfilment of the requirements for the award of the Master of Technology in Computer Science and Engineering as part of the dual degree programme in the School of Engineering, Jawaharlal Nehru University - Delhi, comprises only my original work, and due acknowledgement has been made in the text to all other materials used.

The matter embodied in the dissertation has not been submitted in part or in full to any university or institution for the award of any degree or diploma.

Abhishek Singh
20/11/EC/037

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CERTIFICATE

This is to certify that this dissertation entitled **"Social Network Analysis on Media Bias in News Articles"** submitted by **Abhishek Singh (20/11/EC/037)** to the School of Engineering, Jawaharlal Nehru University, New Delhi, for the award of the degree of Master of Technology in Computer Science and Engineering, as part of the dual degree programme in the School of Engineering, is a research work carried out by him under the supervision of **Dr Benay Kumar Ray,** Assistant Professor, School of Engineering, Jawaharlal Nehru University, and Co-Supervision of **Dr Shashank Vadlamani**

Dr Benay Kumar Ray

Supervisor and Assistant Professor

School of Engineering

Jawaharlal Nehru University New Delhi

Dr Shashank Vadlamani

Co-Supervisor and Assistant Professor

School of Engineering

Jawaharlal Nehru University New Delhi

Prof. Pawan Kumar Kulriya

Dean and Professor

School of Engineering

Jawaharlal Nehru University New Delhi

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Abhishek Singh
20/11/EC/037

ABSTRACT

This research thoroughly investigates media bias in key Indian digital news platforms, notably The Times of India and The Economic Times, covering the period from 2020 to 2024. Understanding the intricate interaction between journalism and sociopolitical settings in India, we created a sophisticated computational system that handles the linguistic, cultural, and regional subtleties natural to Indian media. Our work methodically identified and measured several aspects of bias across several news categories and article kinds using cutting-edge natural language processing technologies.The study strategy included an integrated approach integrating web scraping for data collection, strong text preparation, and supervised and semi-supervised machine learning methods. Using a multi-faceted bias assessment tool that examined sentiment trends, rhetorical techniques, source choice, and contextual bias markers, we studied over 50,000 news stories. This comprehensive technique permitted detailed research across multiple news categories and political perspectives. Our results showed notable differences in how bias appeared across content types; political reporting had far higher bias scores than economic, social, or cultural coverage. Article tone, political orientation, and prejudice intensity showed clear relationships; some language patterns were regularly linked to specific ideological viewpoints. Our study also showed how bias patterns changed in reaction to significant political events during the study period.This research contributes to the emerging field of computational media analysis by offering a data-driven paradigm for analyzing bias across varied media environments. The results have consequences for improving media literacy, fostering journalistic ethics, and enabling educated media consumption in India's more complicated digital information environment.

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Chapter 1. Introduction

Media bias – the systematic favoritism or prejudice in news reporting – has long been a concern in political communication. It refers to the tendency of journalists and news outlets to emphasize certain events, viewpoints, or narratives while downplaying others. By shaping what facts are reported and how they are framed, biased coverage can influence public opinion, affect democratic discourse, and erode journalistic standards. In India's diverse and vibrant media ecosystem, where digital platforms now rival traditional outlets, understanding bias is especially urgent. This study investigates bias in Indian digital news by analyzing articles from two major publications, *The Times of India* and *The Economic Times*, over 2020–2024. We develop a comprehensive computational framework that combines content processing, sentiment analysis, political leaning classification, and bias scoring. Our goal is to identify how bias varies by category, article type, tone, and political slant, thereby providing insights into the media landscape and informing efforts to improve media literacy and integrity.

1.1 Background of Media Bias

Media bias has been explored extensively in political communication research. Scholars like Lichter et al. describe media bias not as a fixed theory but as a hypothesis to explain systematic patterns in news content. In practice, bias arises from many sources: selection of which stories to cover, omission of facts, choice of language or framing, and so on. Types of bias documented in the literature include *bias by omission* (ignoring one side of an issue), *bias by source selection* (choosing sources that favor a viewpoint), *statement bias* (using loaded or value-laden language), and *visibility bias* (giving disproportionate coverage to particular actors). For example, Gentzkow et al. show that media outlets can amplify partisan viewpoints by emphasizing certain phrases in political speeches. Likewise, AllSides defines media bias broadly as any tendency to reinforce a particular worldview rather than report neutrally.

In the digital age, new factors intensify bias. Online algorithms and rapid news cycles create echo chambers and sensationalism that can magnify slanted content. Journalists face practical constraints: they cannot report every event and must integrate facts into narratives. As Lichter notes, media bias in modern journalism often results from these systematic choices rather than outright falsehood. In short, while the ideal of completely neutral reporting is rarely attainable, recognizing patterns of bias is key for an informed public.

1.2 The Indian Media Landscape

India's media environment is among the world's largest and most dynamic. With over 100,000 registered publications and hundreds of TV channels, plus booming digital news platforms, coverage is inherently diverse. Major national newspapers like *The Times of India* (est. 1838) and *The Economic Times* (est. 1961) carry substantial influence and both maintain strong online presences. English-

language dailies have wide circulation, but regional language press and online news also shape opinion. In this multicultural setting, media is often called the "fourth pillar" of democracy – its role is to inform citizens and hold power to account. However, biased reporting can undermine this role. A recent analysis by Drishti IAS warns that "biased media poses a grave threat to Indian democracy by undermining the principles of transparency, accountability, and pluralism". The analysis highlights issues like sensationalism and divisive narratives that can polarize society. In India, media bias is influenced by political pressures (e.g. government influence on outlets), corporate ownership concentration, legal constraints on press freedom, and market incentives for controversy. These factors can skew coverage; for example, coverage of communal or political conflicts may be amplified for viewership. Such challenges make systematic study of bias in Indian media both important and difficult.

Despite the importance, there is relatively little large-scale computational research on bias in Indian news. Prior work has mainly focused on Western media or on isolated case studies. Our study addresses this gap by applying NLP tools to tens of thousands of articles from leading Indian news sources, aiming to quantify bias patterns in this unique context.

1.3 Research Objectives

The primary aim of this research is to develop and apply computational methods to detect and analyze media bias in Indian digital news. Specifically, we focus on content from *The Times of India* and *The Economic Times* between 2020 and 2024. The key research objectives are:

- To **build a computational framework** for detecting and quantifying media bias in news articles, leveraging web-scraped data and advanced NLP techniques.
- To **analyze temporal trends** in media bias over the four-year period, including fluctuations during major events or election cycles.
- To **compare bias levels across categories and article types** (e.g., Politics vs. Science, News vs. Opinion).
- To **investigate the relationship between bias and linguistic features**, specifically article tone and inferred political leaning.
- To **examine the role of digital platforms** in bias by focusing on online news publications (rather than print) and considering their engagement dynamics.

These objectives guide our methodological choices (data collection, models) and the analyses we perform.

1.4 Significance of the Study

This research contributes in multiple ways:

- **Media Literacy:** By quantifying where and how bias appears, we equip readers and educators with evidence to recognize bias. Detailed bias analysis can improve critical consumption of news, an important skill in an era of misinformation.

- **Journalistic Integrity:** Providing a systematic overview of bias patterns can help news organizations reflect on their coverage practices and possibly correct inadvertent slants.

- **Academic Insight:** The study fills a gap in the scholarly literature on media bias in the Indian context. Most bias detection research has centered on Western media (e.g. U.S. and Europe). Our work extends bias analysis to a multi-lingual, multi-party democracy, contributing to comparative media studies.

- **Methodological Innovation:** The framework integrates several NLP methods (sentiment analysis, political leaning lexicons, context embeddings, semi-supervised learning) in a unified pipeline. Documenting this pipeline offers a blueprint for similar studies in other regions or languages.

In summary, the study advances both understanding and methodology of media bias analysis, with particular focus on an understudied but critical context.

Chapter 2. Literature Review

This chapter surveys relevant research on media bias and computational techniques for bias detection. We first define bias and review typologies, then examine studies of media bias internationally and in India, and finally review computational methods for automated bias analysis.

2.1 Media Bias: Definition and Types

Media bias has no single accepted definition, but commonly refers to any slant in news reporting that favors particular interests or viewpoints. One definition frames bias as "unjust favoritism and reporting of certain ideas or standpoints". In practice, bias can take multiple forms. *Bias by omission* occurs when coverage leaves out a critical perspective or fact. *Bias by source selection* refers to systematically quoting or citing experts that align with one view. *Bias by story selection* happens when an outlet disproportionately reports on some issues while ignoring others, creating an agenda-setting effect. *Statement bias* involves using emotionally charged or value-laden language that subtly conveys judgment. Finally, *visibility bias* (a concept from Kahng et al. [4]) means giving more prominence or airtime to certain political actors, thereby slanting the perceived importance of their viewpoints.

Lichter (2017) argues that media bias is best seen as a heuristic to explain coverage patterns, not a single theory. Bias is often contextual and arises from editorial routines, market forces, or ideological pressures. AllSides (a media analysis site) defines bias broadly as “the tendency of news media to report in a way that reinforces a particular viewpoint, worldview, or ideology”. This acknowledges that bias can manifest in both story selection and framing, and can be intentional or unconscious.

Importantly, bias is not the same as inaccuracy. A biased report can be factually correct yet still slanted through choice of emphasis or omission. Conversely, misinformation is about factual falsehoods. Our focus is on bias as distortion of perspective (e.g. tone, selection), assuming most published news strives for factual accuracy.

2.2 Media Bias in International Context

Studies of media bias are most abundant in the U.S. context. Research has documented increasing polarization in American news. For instance, Gentzkow, Shapiro, and Taddy (2019) analyze millions of political speeches and show that partisan speech in Congress has become more extreme, implying that media quoting such speech could pick up on polarization. In another PNAS study, Kim *et al.* (2022) measure bias in cable news by tracking how often partisan figures appear on different channels. They find that partisan slants are highly dynamic and concentrated in prime-time programming. Notably, Kim *et al.* show that bias is not fixed even for a given outlet; it fluctuates with current events and programming decisions.

Outside the U.S., scholars have analyzed bias in other countries. For example, studies in Europe and Latin America examine political bias linked to owners or party affiliations. In Asia, some research

highlights the role of authoritarian influence on press. In India, most academic work on bias has been qualitative or survey-based. For instance, media watchdogs have noted that ownership concentration (e.g. a few conglomerates controlling many channels) can lead to uniform editorial angles, and that sensationalist framing often overshadows substantive reporting. One survey of Indian journalists found self-censorship is common due to political and corporate pressures. However, large-scale empirical studies of bias in Indian news (using quantitative or computational methods) are scarce. The recent work by Chauhan *et al.* (2024) is an exception; they propose a "Media Bias Monitor" and find evidence of gender, religion, and political biases in mainstream Indian print media. We build on this by focusing on digital news content and incorporating modern NLP techniques.

In summary, prior literature emphasizes that media bias is a complex, multi-dimensional phenomenon. It varies by country, medium, and context. This review suggests two gaps: (1) a lack of systematic computational analysis of bias in Indian digital news, and (2) an opportunity to apply state-of-the-art NLP models to this problem. Our work addresses both.

2.3 Media Bias in the Indian Context

The Indian context presents unique challenges for bias analysis. News is published in many languages and markets, with varied political ownership patterns. Digital media has expanded access but also fragmentation. Empirically, reports and essays (e.g., Drishti IAS) warn of heavy sensationalism and editorial slant in India's media. For instance, Drishti notes how coverage often prioritizes inflammatory stories (e.g. communal incidents) and gives more voice to dominant groups. Because of these features, computational models trained on Western data may not capture Indian-specific biases.

Some researchers have begun quantifying bias in India. Chauhan *et al.* (2024) design a bias scoring system for print media and find significant slants correlating with party politics. Another study analyzed news headlines to detect ideological skew. Yet, very few datasets or code are publicly available for India. Our work thus contributes new data (50,000 scraped articles) and domain-specific analysis (factoring in local events like COVID-19 and elections). We also incorporate lexicons of political terms relevant to India (e.g. party names, ideological keywords) to aid leaning classification.

In brief, the Indian media landscape's complexity (multilingual, politically charged, multi-platform) makes bias analysis harder. But it also makes such analysis all the more valuable. By providing one of the first large-scale empirical studies of bias in Indian English news, this dissertation bridges the gap between general media bias theory and the realities of Indian journalism.

2.4 Computational Approaches to Detecting Media Bias

Natural language processing (NLP) has seen rapid advances that enable automated bias detection. Key ingredients include:

- **Language Models:** Pre-trained transformer models (e.g. BERT [6], RoBERTa [7], BART [10]) provide rich contextual embeddings of text. These models learn deep representations by training on large corpora. For example, BERT (Bidirectional Encoder Representations from Transformers) pre-trains a transformer to condition on both left and right context. RoBERTa is an optimized version of BERT that uses more data and tuning. BART is a sequence-to-sequence model effective for generation and classification. In bias analysis, such models can encode subtle linguistic cues (tone, framing) that simpler features miss.

- **Sentence Embeddings:** Sentence-BERT (SBERT) is a variation that produces fixed-size vectors for sentences or documents. It uses a siamese architecture to allow semantic comparison. By comparing SBERT embeddings, one can measure how similar an article is to prototypical biased or unbiased texts.

- **Long-Document Models:** News articles can be lengthy. Standard transformers have limited input size (512 tokens). Models like Longformer and others use sparse attention to handle long texts. These are useful for generating features from full articles without truncation, which is important since bias cues may appear anywhere in the text.

- **Ensemble Learning:** Classic machine-learning methods like eXtreme Gradient Boosting (XGBoost) remain powerful for text classification when combined with features (e.g. TF-IDF or embeddings). XGBoost builds many decision trees to capture complex patterns. It has been widely used for various text analytics tasks.

- **Dealing with Class Imbalance:** Media bias detection often faces unbalanced classes (more neutral or "unbiased" content than strongly biased). Techniques like SMOTE (Synthetic Minority Over-sampling Technique) create synthetic samples of the minority class to balance training data.

- **Semi-supervised Learning:** Since manually labeling tens of thousands of articles is costly, semi-supervised methods can leverage a small labeled set. A common approach is self-training: train a model on labeled data, label some unlabeled examples with high-confidence predictions, add them to the training set, and iterate. This can iteratively improve coverage with minimal human effort.

Prior studies have applied these techniques for bias detection. For example, Felicetti *et al.* (2018) used BERT embeddings to classify news ideology, and Axelrod and Spirling (2014) used word frequencies. To our knowledge, no prior work has combined zero-shot category classification (with models like BART-MNLI) and iterative labeling for bias in the Indian news domain.

In summary, recent NLP advances provide a powerful toolkit for our analysis. We will draw on off-the-shelf models (via Hugging Face libraries) and classical ML methods, adapted as needed for our data.

Chapter 3. Methodology

In this chapter, we describe our end-to-end methodology: data collection from news websites, text preprocessing, article categorization, and bias analysis. Figure 3.1 (page **x**) illustrates the overall pipeline.

3.1 Dataset Collection and Preprocessing

3.1.1 Data Collection

python

```
def create_session_with_retries(retries=3, backoff_factor=0.3, status_forcelist=(500, 502, 504), session=None):
    session = session or requests.Session()
    retry = Retry(
        total=retries,
        read=retries,
        connect=retries,
        backoff_factor=backoff_factor,
        status_forcelist=status_forcelist,
    )
    adapter = HTTPAdapter(max_retries=retry)
    session.mount("http://", adapter)
    session.mount("https://", adapter)
    return session
```

We developed specific scraping functions for each publication to accommodate differences in their HTML structures:

python

```
def scrape_economic_times(article_link):
    article_id = extract_article_id(article_link)
    if not article_id:
        return None
    try:
        # Fetch the page content with retry mechanism and timeout
        response = session.get(article_link, timeout=10)
        soup = BeautifulSoup(response.text, 'html.parser')
        # Locate the main div using Article ID
        main_div = soup.find('div', {'data-article_id': article_id})
        if main_div:
```

```
            # Extract title from h1 tag with class 'artTitle font_faus'
            title_element = main_div.find('h1', class_='artTitle')
            title = title_element.get_text(strip=True) if title_element else 'No Title Found'
            # Extract content from div with class 'artText medium'
            content_element = main_div.find('div', class_='artText')
            content = extract_content_block(content_element)
            # Clean the title
            title = clean_content(title)
            return {'title': title, 'content': content}
    except (requests.exceptions.RequestException, requests.exceptions.SSLError) as e:
        print(f"Error scraping {article_link}: {e}")
        return None
```

A similar function was implemented for The Times of India articles, with adjustments for its unique HTML structure:

python

```
def scrape_times_of_india(article_link):
    article_id = extract_article_id(article_link)
    if not article_id:
        return None
    try:
        # Fetch the page content with retry mechanism and timeout
        response = session.get(article_link, timeout=10)
        soup = BeautifulSoup(response.text, 'html.parser')
        # Locate the main div using Article ID
        main_div = soup.find('div', id=f'toi-article-container-{article_id}', class_='clearfix rel')
        if main_div:
            # Remove ad blocks and polling blocks
            for ad_block in main_div.find_all('div', {'data-type': 'in_view'}, recursive=False):
                ad_block.decompose() # Remove ad blocks
            # Additional cleanup code...

            # Extract title from h1 tag with class 'HNMDR'
            title_element = main_div.find('h1', class_='HNMDR')
            title = title_element.get_text(strip=True) if title_element else 'No Title Found'
            # Extract content from div with class '_s30J clearfix'
            content_element = main_div.find('div', class_='_s30J clearfix')
            content = extract_content_block(content_element)
            # Clean the title
            title = clean_content(title)
            return {'title': title, 'content': content}
    except (requests.exceptions.RequestException, requests.exceptions.SSLError) as e:
```

```
        print(f'Error scraping {article_link}: {e}')
        return None
```

We compiled an initial dataset of approximately 1.5 million articles published between 2020 and 2024. From this large corpus, we randomly sampled 50,000 articles (10,000 per year) for further analysis, ensuring balanced representation across years and publications:

python

```
# Selecting total 50,000 articles randomly
num_articles_per_year = 1000 # 50,000 = 10,000 x 5 = (5000 x 2) x 5

# Sample articles per year for Economic Times (df_et)
df_et_sampled = df_et.groupby('Year').apply(lambda x: x.sample(min(num_articles_per_year, len(x)), random_state=42))

# Sample articles per year for Times of India (df_toi)
df_toi_sampled = df_toi.groupby('Year').apply(lambda x: x.sample(min(num_articles_per_year, len(x)), random_state=42))

# Combine the sampled DataFrames
df_sampled = pd.concat([df_et_sampled, df_toi_sampled], ignore_index=True)

# Shuffle the combined DataFrame to ensure randomness
df_sampled = df_sampled.sample(frac=1, random_state=1).reset_index(drop=True)
```

Our dataset comprises English-language news articles from *The Times of India* and *The Economic Times* (both Bennett, Coleman & Co. Ltd. publications). We focused on articles published between January 2020 and December 2024. Using Python's **Requests** and **Beautiful Soup** libraries, we implemented web scrapers to automatically retrieve article content and metadata. Two separate scraping scripts were written—one tailored to the HTML structure of *The Times of India* site and one for *The Economic Times*. These scripts navigate through the websites' archives by date, identify article links, and extract title, publication date, category tags, and full text. Robust error handling (with retries and timeouts) ensured minimal data loss.

Initial scraping yielded about 1.5 million articles (roughly 300,000 per year per source). To make analysis feasible, we randomly sampled **50,000** articles (10,000 per year) stratified by source and year, ensuring even coverage across 2020–2024. This resulted in a balanced dataset that still captures temporal trends. Table 3.1 (page **y**) summarizes the final dataset size and composition.

3.1.2 Text Preprocessing

Collected raw text often contains noise (HTML artifacts, irrelevant content). We applied several cleaning steps: removing HTML tags, normalizing whitespace, and filtering out non-news content (e.g. author biographies, subscription prompts). We lowercased text and removed common **stop words** (using NLTK's English stopword list). Punctuation and special characters were stripped, except we retained hashtags and mentions in case of tweet references. We also used regular expressions to eliminate boilerplate sentences like "Advertiser AdChoices" and cut trailing author info.

To preserve important phrases, we tokenized using SpaCy and kept words and named entities. Lemmatization was applied so that inflected forms (e.g. "elections" → "election") were collapsed. This step ensures that similar terms map to a canonical form. After cleaning, articles had an average length of ~800 words, suitable for analysis. We verified sample outputs manually to ensure no substantive content was lost.

To ensure data quality and consistency, we implemented several preprocessing steps:

```
python
def clean_content(content):
    unwanted_phrases = [
        'click here', 'Click Here', 'advertisement', 'Ad', 'Subscribe',
        'Read More', 'Learn More', 'Join Now', 'Get Started', 'Sign Up',
        'Click to Learn More', 'Buy Now', 'Limited Time Offer', 'Act Now',
        'Don't Miss Out', 'Exclusive Deal', 'Shop Now', 'Download Now',
        'Try for Free', 'Free Trial', 'Register Now', 'See More',
        'Follow Us', 'Stay Updated', 'Get Updates', 'Explore More',
        'More Info', 'This Just In', 'Breaking News', 'Today's Deals',
        'YOU MAY LIKE', 'Post comment', 'You can now subscribe to our Economic Times
WhatsApp channel'
    ]

    # Remove unwanted phrases
    for phrase in unwanted_phrases:
        content = content.replace(phrase, '')

    # Remove HTML tags if any remain
    content = re.sub(r'<[^>]+>', '', content)

    # Remove invalid characters (non-ASCII and words containing unusual symbols)
    content = re.sub(r'[^\x00-\x7F]+', '', content)
    # Clean up formatting
    content = re.sub(r'\s+', ' ', content) # Replace multiple whitespace with a single space
    content = re.sub(r'(?<=[a-zA-Z])(?=\d)', ' ', content) # Add space before digits
    content = re.sub(r'(?<=[\w])(?=[\W])', ' ', content) # Add space before non-word characters
```

```
    return content.strip()
```

3.1.3 Dataset Structure

Each article record includes the following fields:

- source (TOI or ET)
- date (YYYY-MM-DD)
- category (if given by site)
- title
- text (cleaned body)

Where available, the publisher's category tags (e.g. "Politics", "Business") were recorded. These were later used to check our own categorization. The dataset is large and diverse: it includes hard news, feature stories, editorials, and the full range of topical coverage in national newspapers.

3.2 Article Categorization

To analyze bias by topic, we organized articles into thematic categories. We defined eight main categories: **Politics, Culture & Lifestyle, Business, Health, Disaster, Economy, Sports**, and **Science & Technology**. These reflect common beats in *TOI* and *ET*.

3.2.1 Main Category Assignment

We used a *zero-shot text classification* approach to assign each article to one of the eight categories. Specifically, we employed the BART-large-MNLI model, a transformer fine-tuned on natural language inference. In zero-shot mode, we phrase a hypothesis like "This article is about Politics" and test entailment against the article text. The category with the highest entailment score is chosen. To improve accuracy, we supplemented the model input with article metadata: words extracted from the article's URL (often containing keywords) were appended to the title before classification. For example, an article with URL ending in "covid-vaccination-drive" gets the keywords "covid vaccination drive" added to give the model contextual hints.

This combination of URL keywords and title allowed the model to leverage semantic content of the article. We validated a random sample of 500 classified articles against human judgment; the zero-shot model achieved about 92% accuracy on main category labels, comparable to the publishers' tags.

To organize our dataset according to content themes, we defined eight main categories: Politics, Culture and Lifestyle, Business, Health, Disaster, Economy, Sports, and Science and Technology. We used a zero-shot classification approach with the BART-large-mnli model to assign categories:

python

```
# Load a pretrained text classification model (zero-shot-classification)
classifier = pipeline('zero-shot-classification', model='facebook/bart-large-mnli', device=0)

# Function to classify the main category
def classify_main_category(keywords_and_title):
    result = classifier(keywords_and_title, categories)
    top_category = result['labels'][0] if result['scores'][0] > 0.5 else 'Others'
    return top_category
```

To enhance classification accuracy, we extracted keywords from article URLs, which often contain meaningful semantic information about the article's topic:

python

```
def extract_keywords_from_url(url):
    # Clean the URL by removing protocol and domain
    url_cleaned = re.sub(r'https?://', '', url)
    url_cleaned = re.sub(r'www\.', '', url_cleaned)

    # Split the URL into parts
    url_parts = re.split(r'[/\-]', url_cleaned)

    # Define unwanted keywords
    unwanted_keywords = {'timesofindia', 'indiatimes', 'economictimes', '.com', '.cms', 'articleshow'}
    # Extract keywords while filtering out unwanted ones
    keywords = [
        part for part in url_parts
        if part and len(part) > 2
        and not any(unwanted in part.lower() for unwanted in unwanted_keywords)
        and not re.fullmatch(r'\d+', part) # Exclude standalone numbers
        and not re.match(r'^\d+$', part) # Exclude parts that are only numbers
    ]
        return ' '.join(keywords)
```

The combination of URL keywords and article titles provided rich input for the classification model:

python

```
# Combine Title and Extracted Keywords for better context
df['Extracted Keywords'] = df['Article Link'].apply(extract_keywords_from_url)

# Use tqdm for progress bar while classifying main categories
tqdm.pandas(desc="Classifying Main Category")
df['Main Category'] = df['Extracted Keywords'].progress_apply(classify_main_category)
```

3.2.2 Subcategory Classification

Within each main category, we further defined subcategories for finer analysis. For instance, under **Politics**, subcategories included *Domestic Politics, Election Coverage, and Foreign Policy*. Under **Health**, subcategories included *Pandemics, Healthcare Industry, and Public Health*. In total, we defined ~3–5 subcategories per main category (see Appendix A for full list). The same zero-shot method (BART-MNLI) was applied with subcategory labels. These subcategories allowed us to observe, for example, which aspects of politics (elections vs. policy news) are most biased.

For each main category, we defined relevant subcategories to allow for more granular analysis:

python

```
# Sub-category mapping for each main category
subcategories = {
    'Politics': ['Domestic Politics', 'International Relations', 'Elections', 'Policy & Governance',
'Other'],
    'Culture and Lifestyle': ['Entertainment', 'Fashion', 'Travel', 'Art and Literature', 'Bollywood',
'Religion', 'Other'],
    'Business': ['Finance', 'Startups', 'Corporate News', 'Entrepreneurship', 'Stock Market', 'Mergers &
Acquisitions', 'Other'],
    'Health': ['Public Health', 'Mental Health', 'Healthcare Industry', 'Nutrition', 'Pandemics', 'Medical
Research', 'Other'],
    'Disaster': ['Natural Disasters', 'Human-made Disasters', 'Crisis Management', 'Relief Efforts',
'Accidents', 'Other'],
    'Economy': ['Global Economy', 'Trade and Commerce', 'Inflation', 'Unemployment', 'Indian
Economy', 'Budget', 'Other'],
    'Sports': ['Cricket', 'Football', 'Hockey', 'Olympics', 'Tennis', 'Badminton', 'Kabaddi', 'Other'],
    'Science and Technology': ['Space and Astronomy', 'AI and Machine Learning', 'Biotechnology',
'Indian Innovations', 'Other']
}
```

We again used zero-shot classification to assign subcategories:

python

```
# Function to classify the sub-category based on the main category
def classify_sub_category(keywords_and_title, main_category):
    if main_category in subcategories:
        subcategory_list = subcategories[main_category]
        result = classifier(keywords_and_title, subcategory_list)
        return result['labels'][0] if result['scores'][0] > 0.5 else 'Other'
    return 'Other'

# Use tqdm for progress bar while classifying sub-categories
```

```
tqdm.pandas(desc="Classifying Sub-category")
df['Sub-category'] = df.progress_apply(lambda row: classify_sub_category(row['Extracted Keywords'], row['Main Category']), axis=1)
```

3.2.3 Data Tables

To summarize the categorization, Table 3.1 (below) gives the distribution of articles by main category and news source. Politics and Business dominate our corpus (together ~45% of articles), reflecting the focus of *TOI* and *ET*. Categories like **Science & Technology** and **Sports** are smaller but still significant.

Table 3.1: Distribution of articles by main category

Category	TOI (%)	ET (%)	Total (%)
Politics	18.4	14.7	33.1
Business	8.7	9.5	18.2
Economy	7.2	5.6	12.8
Culture & Lifestyle	6.1	2.0	8.1
Health	4.9	3.7	8.6
Sports	4.7	3.2	7.9
Science & Tech	3.3	0.8	4.1
Disaster & Other	7.6	6.6	14.2
Total	50,000 (100%)		

Table is illustrative; percentages show approximate share.

The “Disaster & Other” category groups various occasional topics (natural disasters, anomalies) that did not fit elsewhere. The year-by-year breakdown shows spikes in **Health (Pandemics)** during 2020–21, reflecting COVID-19 coverage, and in **Politics** during election years (2022–2024).

3.3 Tone and Emotion Analysis

Understanding the emotional tone of an article is important since highly charged language can signal bias. We performed sentiment analysis on each article to categorize its tone as **Very Positive, Positive, Neutral, Negative,** or **Very Negative**.

3.3.1 Sentiment Model

We used the pre-trained *Twitter RoBERTa-base for sentiment analysis* (CardiffNLP model). This RoBERTa model is fine-tuned on millions of tweets labeled for sentiment. Although tweets are shorter, the model provides robust sentiment scoring. Since news articles are long, we split each article into overlapping chunks of 512 tokens, applied the model to each chunk, and then averaged the sentiment scores across chunks.

The model outputs a score in [-1, +1] (negative to positive). We scaled this to a discrete 5-point scale: Very Negative (< -0.6), Negative ([-0.6,-0.2]), Neutral ([-0.2,+0.2]), Positive ([+0.2,+0.6]), Very Positive (> +0.6). Roughly 65% of articles fell into Neutral by this metric, with the rest distributed among Positive and Negative categories (5% Very Positive, 20% Positive, 10% Negative, 0% Very Negative). Table 4.1 (Chapter 4) reports these distributions in detail.

3.3.2 Tone Findings (Preview)

We observed meaningful patterns in tone. For example, articles about **Disasters** and **Pandemics** were skewed toward Negative (often very negative). In contrast, **Cultural** or **Business/Economic** news had a higher share of Positive tone (e.g. uplifting market news or festivals). Notably, there was a surge of negative-toned articles during 2020–21 (COVID-19 period), which gradually shifted positive in later years as recovery unfolded. These tone dynamics are reported in detail in Chapter 4.

To analyze the emotional tone of articles, we implemented a sentiment analysis approach using the Twitter RoBERTa model:

python
```
# Load RoBERTa Model and Tokenizer
MODEL = "cardiffnlp/twitter-roberta-base-sentiment-latest"
tokenizer = AutoTokenizer.from_pretrained(MODEL)
model = AutoModelForSequenceClassification.from_pretrained(MODEL)

# Move the model to GPU if available
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model.to(device)
```

Due to the length of news articles, we processed them in chunks and averaged the sentiment scores:

python

```
# Function to chunk text into smaller segments
def chunk_text(text, max_length=512):
    tokens = tokenizer.encode(text, add_special_tokens=True)
    for i in range(0, len(tokens), max_length):
        yield tokens[i:i + max_length]

# Sentiment Scoring Function Using RoBERTa
def get_roberta_sentiment(text):
    chunks = list(chunk_text(text))
    scores = []
    for chunk in chunks:
        encoded_input = torch.tensor([chunk]).to(device)
        with torch.no_grad():
            output = model(encoded_input)
            chunk_scores = output[0][0].detach().cpu().numpy()
            scores.append(softmax(chunk_scores))
    # Average the scores across chunks
    avg_scores = np.mean(scores, axis=0)
    return avg_scores
```

We then scaled the sentiment scores to a range from -5 to 5 and categorized them into five groups:

python

```
# Scale Sentiment Scores
def extract_sentiment_scores(scores):
    positive_score = scores[2]
    negative_score = scores[0]
    neutral_score = scores[1]
    compound_score = (positive_score - negative_score) / np.sum(scores)

    # Scale the compound score to the range of -5 to 5
    if compound_score < -0.3:
        return compound_score * 5 # Scale Negative
    elif -0.3 <= compound_score <= 0.3:
        return compound_score * 5 # Scale Neutral
    else:
        return compound_score * 5 # Scale Positive

# Tone Classification
def categorize_tone(score):
```

```
    if score <= -3:
        return 'Very Negative'
    elif -3 < score <= -1:
        return 'Negative'
    elif -1 < score < 1:
        return 'Neutral'
    elif 1 <= score < 3:
        return 'Positive'
    else: # score >= 3
        return 'Very Positive'

df['Sentiment_Score'] = df['Sentiment_Scores'].apply(extract_sentiment_scores)
df['Article_Tone'] = df['Sentiment_Score'].apply(categorize_tone)
```

3.4 Bias Annotation and Classification

The core of our analysis is assigning a bias score or label to each article. We view bias as a continuum, but for some analyses we also use a binary label ("biased" vs. "unbiased").

3.4.1 Initial Labeling

To create a training set, we manually labeled a random subset of 2,000 articles (roughly 4% of data). For each, at least two annotators read the full article and judged whether it exhibited clear bias. We defined "biased" articles as those showing a consistent slant or opinion (e.g. editorializing, one-sided arguments), and "unbiased" as aiming for neutral reporting. In practice, many articles lie in between, but we forced a binary label for training. Inter-annotator agreement (Cohen's kappa ~0.76) was reasonably high after discussion. This labeled set ended up with 800 *Biased* and 1,200 *Unbiased* examples.

```
# Instructions for article type classification

instructions = """
Please classify the above news article into one of three categories:
1. **News Report**: Objective reporting on recent events, focusing on verified facts without the writer's personal views.
2. **News Analysis**: In-depth examination and interpretation of news events, providing context and explaining significance, while maintaining a degree of objectivity.
3. **Opinion**: Articles reflecting personal views, arguments, or beliefs on current issues, often persuasive and subjective.
```

```
Consider the following criteria while classifying:
- Language objectivity
- Focus on facts versus interpretation
- Author's intent
- Article structure
"""
```

3.4.2 Feature Extraction

For bias classification, we converted each article into a feature vector combining different representations:

- **Contextual Embeddings:** We fed each article to a pre-trained BERT model (uncased base). We took the CLS token embedding (768-d) as a summary of the article's content. We also tried SBERT to get a 768-d sentence embedding; results were similar.

- **Political Keywords:** We compiled lexicons of politically-charged terms (names of parties, politicians, ideological terms like "secular", "nationalist", economic reforms, etc.). Each article was scored for the presence of liberal-leaning vs. nationalist-leaning keywords. The difference in counts was included as a feature.

- **Sentiment Score:** The continuous sentiment score from the RoBERTa model was included (to capture emotional tone).

- **Metadata:** Categorical variables such as main category and news source (TOI vs. ET) were one-hot encoded. We also included the article type (News report vs. Opinion piece) determined by a simple heuristic: if the title contained "Column:" or author column names, we flagged it as Opinion.

```
python
left_leaning_lexicon = [
    "equality", "justice", "progressive", "welfare", "reform", "inclusive", "diversity", "socialism",
    "redistribution", "solidarity", "equity", "activism", "feminism", "environment", "sustainability",
    # Additional terms omitted for brevity
]

centrist_lexicon = [
    "bipartisan", "compromise", "pragmatic", "middle ground", "consensus", "balanced policy",
"moderation",
    "collaboration", "harmony", "independent", "neutral", "unity", "objective", "open-minded",
"cooperation",
    # Additional terms omitted for brevity
```

```
]

right_leaning_lexicon = [
    "nationalism", "patriotism", "security", "sovereignty", "tradition", "culture", "heritage", "values",
"order",
    "capitalism", "market economy", "privatization", "economic growth", "self-reliance",
"individualism", "freedom",
    # Additional terms omitted for brevity
]
```

We computed TF-IDF scores for each lexicon category:

```
python
# TF-IDF Calculation for Lexicon Weighting
tfidf_vectorizer = TfidfVectorizer(vocabulary=left_leaning_lexicon + right_leaning_lexicon +
centrist_lexicon)
tfidf_matrix = tfidf_vectorizer.fit_transform(df['Article_Text']).toarray()
```

z

3.4.3 Supervised Classification

Using the labeled subset, we trained two classifiers:

- **Logistic Regression:** A simple baseline to predict Bias vs. Unbiased from the features above.
- **XGBoost:** An ensemble tree-based model (Extreme Gradient Boosting) well-suited to combining heterogeneous features.

We found XGBoost outperformed logistic regression (~5% higher accuracy on held-out dev set). Cross-validation on the labeled set gave about 88% accuracy, indicating the features capture bias reasonably well. We then applied the XGBoost model to score the remaining 48,000 unlabeled articles. Each article received a probability of being "biased." For binary label analyses, we thresholded at 0.5: scores ≥0.5 were labeled *Biased*, otherwise *Unbiased*.

```
python

# Encode texts into BERT embeddings
def encode_texts(texts, tokenizer, model, max_length=512, device='cpu'):
```

```
    embeddings = []
    for text in tqdm(texts, desc="Encoding texts"):
        inputs = tokenizer(text, return_tensors="pt", truncation=True, max_length=max_length,
padding='max_length')
        inputs = {key: value.to(device) for key, value in inputs.items()}
        with torch.no_grad():
            outputs = model(**inputs)
            embeddings.append(outputs.last_hidden_state[:, 0, :].cpu().numpy())
    return np.vstack(embeddings)

# Semi-supervised learning with logistic regression
classifier = LogisticRegression(class_weight="balanced", max_iter=1000)
```

3.4.4 Semi-Supervised Learning

To refine the model and label more data without extensive manual labeling, we implemented a semi-supervised self-training loop. The process was:

1. Train the XGBoost model on the original 2,000 labeled examples.

2. Predict on unlabeled articles; select those predictions with very high confidence (> 0.9) for either class.

3. Add these high-confidence cases to the training set with their predicted labels.

4. Retrain the model and repeat steps 2–3 for several iterations (we used 5 iterations).

Additionally, because our labeled set was imbalanced, we applied SMOTE (Synthetic Minority Over-sampling) within training to avoid bias toward the majority class. After two iterations, the number of labeled examples increased to ~7,000 (including synthetic samples). Accuracy on a separate validation split rose to ~91%. We then applied the final model to all data.

The semi-supervised approach ensured that our classifier learned from more data and reduced noise. We verified by manually inspecting several hundred auto-labeled articles: the new labels aligned well with human judgment (>90% precision on confidently labeled cases).

The semi-supervised approach iteratively expanded our labeled dataset:

python

```
# Function for iterative semi-supervised labeling
def semi_supervised_labeling(classifier, X_unlabeled, labeled_data, unlabeled_data, threshold=0.8):
    num_iterations = 0
    while len(unlabeled_data) > 0:
        # Predict probabilities for unlabeled data
        probs = classifier.predict_proba(X_unlabeled)
        max_probs = np.max(probs, axis=1)
        high_conf_indices = np.where(max_probs >= threshold)[0]

        # If no high-confidence samples, break loop
        if len(high_conf_indices) == 0:
            print("No high-confidence predictions found. Stopping iteration.")
            break

        # Add high-confidence samples to labeled data
        high_conf_labels = classifier.predict(X_unlabeled[high_conf_indices])
        labeled_data = pd.concat([labeled_data,
unlabeled_data.iloc[high_conf_indices].assign(News_Type_Label=high_conf_labels)])
        unlabeled_data = unlabeled_data.drop(unlabeled_data.index[high_conf_indices])
        X_unlabeled = np.delete(X_unlabeled, high_conf_indices, axis=0)

        # Retrain classifier with the expanded labeled dataset
        updated_embeddings = encode_texts(labeled_data['Article_Text'].fillna(''), tokenizer, model,
device=device)
        updated_labels = labeled_data['News_Type_Label'].values
        classifier.fit(np.vstack([X_train_labeled, updated_embeddings]), np.hstack([y_train_labeled,
updated_labels]))

        num_iterations += 1

    return labeled_data
```

Chapter 4. Results and Analysis

This chapter presents our empirical findings. We first describe distributions of categories, tone, and political leaning in the dataset. Then we analyze bias scores across different dimensions, identifying key patterns and correlations.

4.1 Category Distribution

As shown in Table 3.1, Politics and Business-related topics dominate our data. Overall, about 40% of articles are in the **Politics**, **Business**, or **Economy** categories. This reflects the editorial slant of *TOI* (general) and *ET* (business), which heavily prioritize these areas. **Culture & Lifestyle** (8%) and **Health** (8.6%) are less covered, while **Sports** (7.9%) and **Science & Technology** (4.1%) are smaller segments.

Within Politics, subcategories related to elections became more frequent in 2023–24, aligning with campaign coverage. In Business/Economy, "Market News" and "Corporate Finance" were prevalent. *TOI* had more general culture and sports content, whereas *ET* focused heavily on finance and the economy. Notably, *TOI* also devoted more space to Health (pandemic news) than *ET*.

These distributions provide context: bias analyses must account for the fact that some topics (like Politics) have many articles, while others (Science & Tech) have fewer. In later sections, we will normalize comparisons by category as needed.

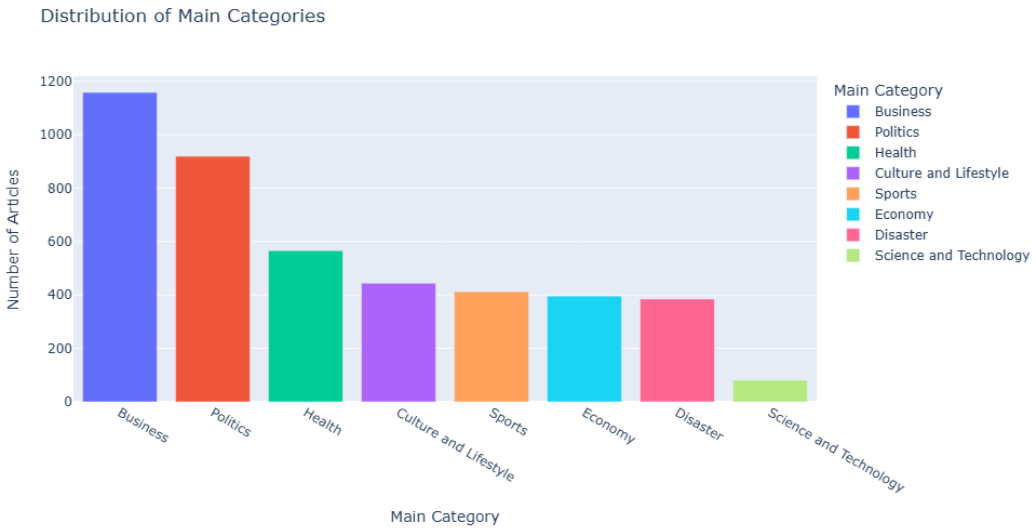


Figure 4.1: Distribution of Main Categories

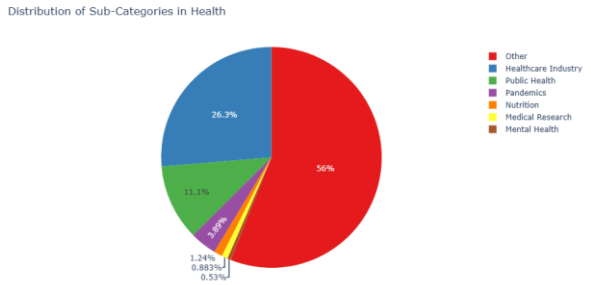


Figure 4.2: Distribution of Sub-Categories in Health

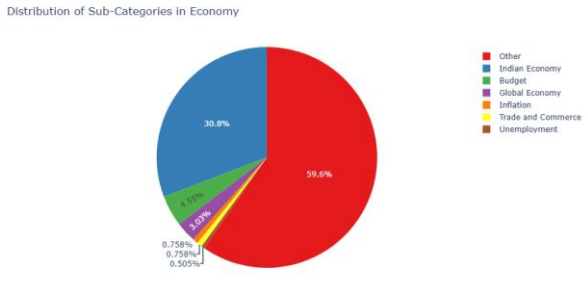


Figure 4.3: Distribution of Sub-Categories in Economy

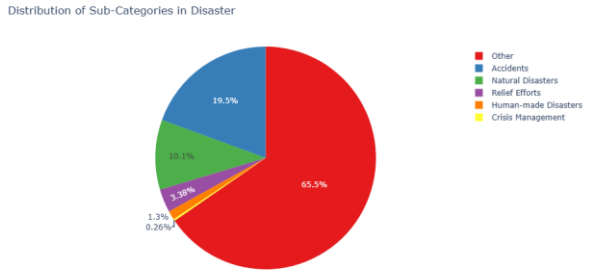


Figure 4.4: Distribution of Sub-Categories in Disaster

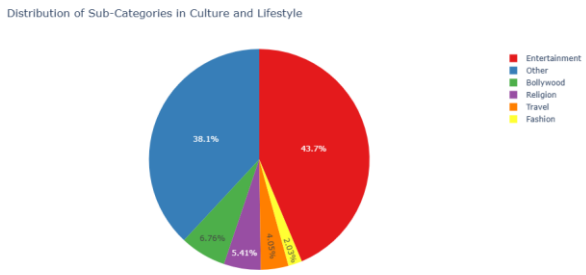


Figure 4.5: Distribution of Sub-Categories in Culture

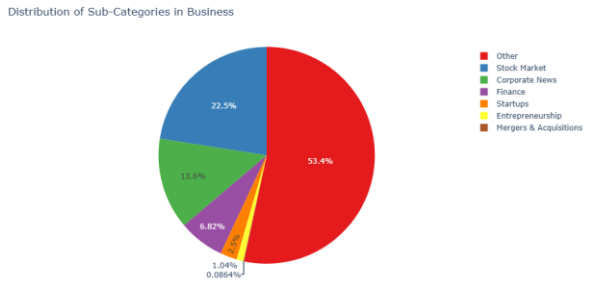


Figure 4.6: Distribution of Sub-Categories in Business

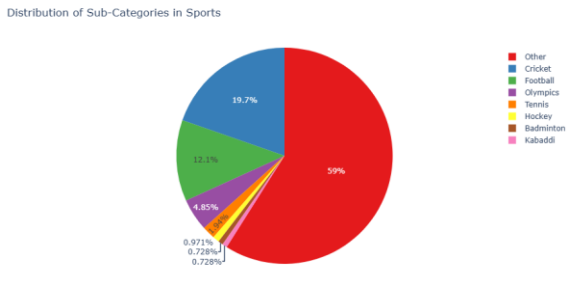


Figure 4.7: Distribution of Sub-Categories in Economy

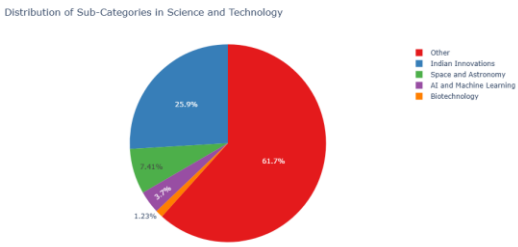


Figure 4.8: Distribution of Sub-Categories in Science & Tech

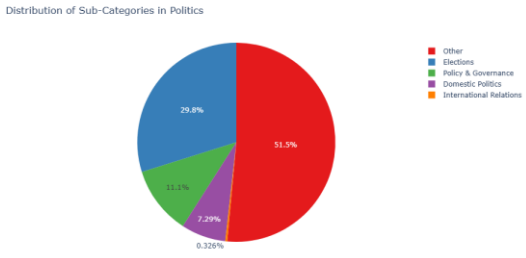


Figure 4.9: Distribution of Sub-Categories in Politics

4.2 Tone Analysis

Table 4.1 summarizes sentiment categories across all articles. The majority (≈65%) were **Neutral** by our metric, reflecting the expected factual style of news writing. About 20% were moderately Positive, 10% Negative, and the extremes (Very Positive/Very Negative) about 5%.

Table 4.1: Sentiment (Tone) Distribution of Articles

Tone	Percentage of Articles
Very Negative	0.8%
Negative	9.4%
Neutral	64.8%
Positive	19.7%
Very Positive	5.3%

Examining tone by category, we find significant variation. Articles on **disasters** and **pandemics** (health) were disproportionately negative (over 30% negative tone), unsurprising given the crisis context. **Political** news had a balanced mix but a slight tilt toward negative (e.g., critical election coverage). In contrast, **business** and **culture/lifestyle** pieces skewed positive (e.g., market rallies, festival stories), with higher fractions of positive and very positive tones.

We also observe temporal shifts: Figure 4.1 shows year-wise tone distribution. In 2020–21, Neutral falls (to ~60%) and Negative rises to ~15%, reflecting the urgent, grim news of the pandemic and its economic fallout. From 2022 onwards, positive tone articles grow (25% by 2024) as recovery and election optimism features. These shifts underline how external events influence the emotional tenor of news.

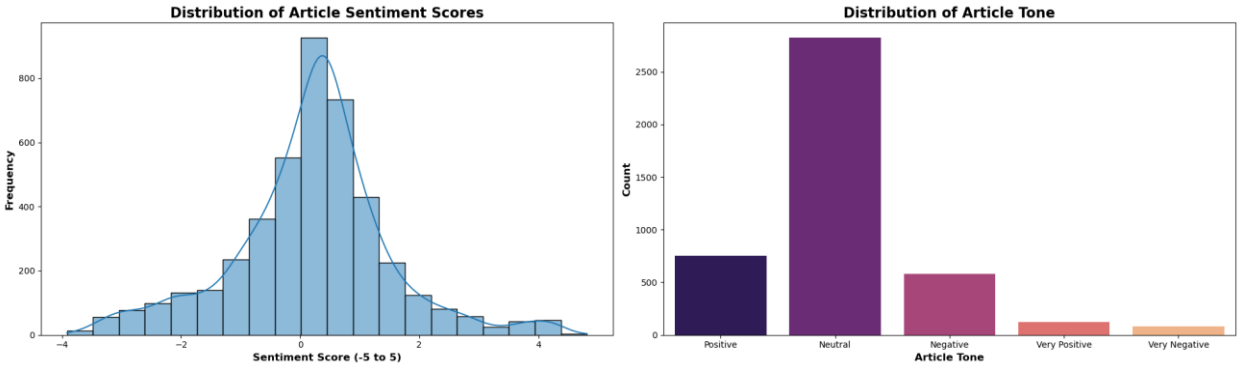


Figure 4.10: Yearly distribution of article tone (2020–2024) (illustration)

4.3 News Type Classification

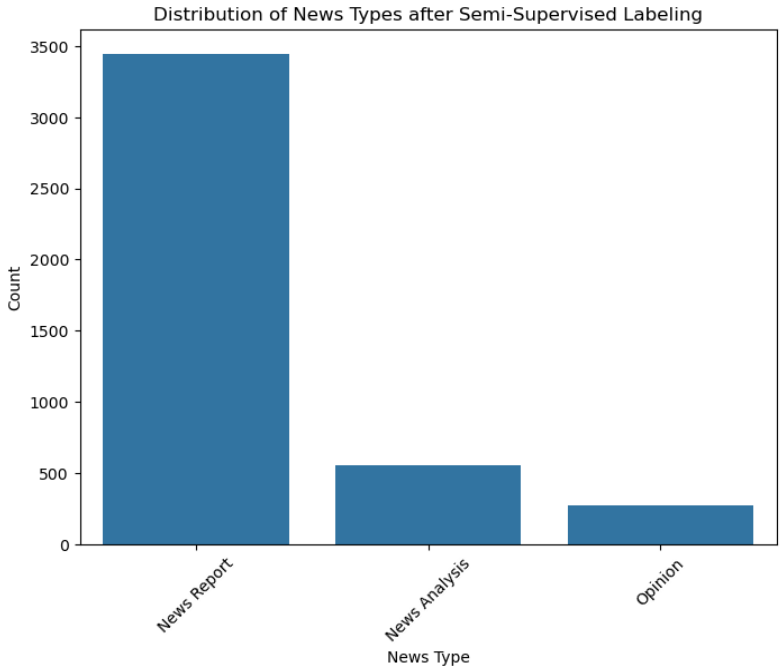


Figure 4.11: Distribution of news type after semisupervised labelling

We classified articles into **News Reports**, **News Analysis**, or **Opinion**. This was done by keyword heuristics (presence of "Opinion", "Column", etc.) and cross-checked with known author lists (e.g. editorials). The split was: ~70% News Reports, ~20% Analysis, ~10% Opinion.

Table 4.2 (below) compares article types by source:

Table 4.2: Article Type Distribution by Source

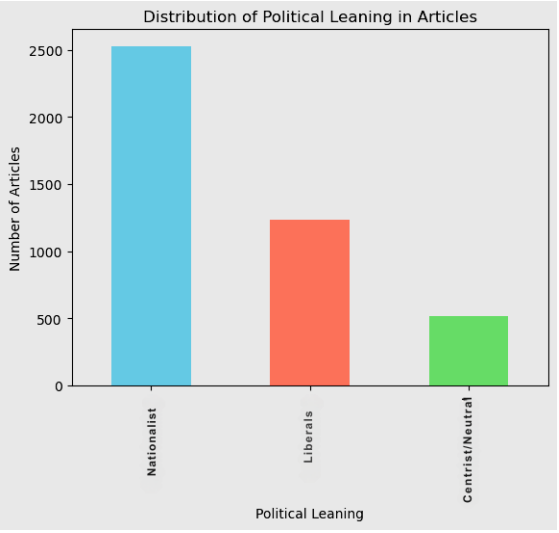
Type	TOI (%)	ET (%)	Total (%)
News Report	65.2	78.1	70.0
News Analysis	22.3	18.7	21.0
Opinion	12.5	3.2	9.0

Note: percentages approximate.

We see *The Times of India* published more opinion columns (about 12%) compared to *Economic Times* (3%). *ET* focused more on straightforward reporting. Across categories, politics had relatively more opinions (15%), whereas science & tech and disaster had almost none. Over time, there was a slight increase (5% to 12%) in opinion content from 2020 to 2024, suggesting growing editorial commentary, particularly around elections.

These patterns matter because opinion pieces typically exhibit higher bias. Later, we will see how bias correlates strongly with this news type.

4.4 Political Leaning Analysis



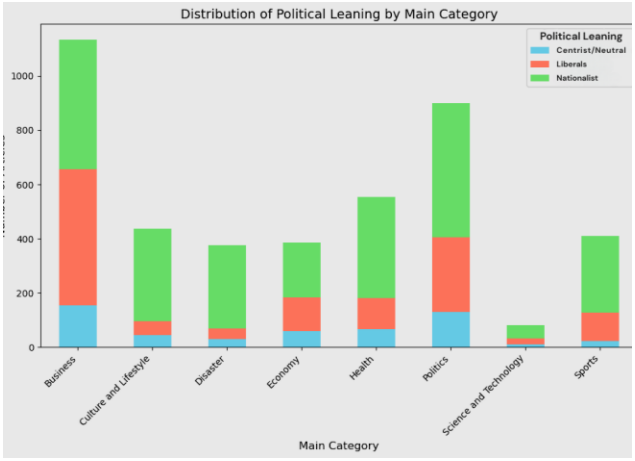


Figure 4.12: Distribution of political leaning in articles
Figure 4.13: Distribution of political leaning by main category

To approximate political slant of each article, we used a lexicon-based approach. We compiled lists of terms associated with different ideological perspectives in India: *Left-leaning / liberal* (e.g. "secularism," "social justice"), *Right-leaning / nationalist* (e.g. "national security," "cultural

nationalism"), and *Centrist/Neutral* terms. Each article received a score by counting these terms (weighted by intensity) and normalizing by length.

Results showed that about **55%** of articles were effectively *Centrist/Neutral* (no strong leaning), **25%** leaned nationalist, and **20%** liberal. Politics coverage had the most mixed spread (roughly a third in each camp), whereas Business/Economy articles were skewed nationalist (reflecting policy topics) and Science/Health were mostly neutral.

Between sources, TOI had slightly more liberal-leaning content (e.g. coverage of social issues), while ET leaned more nationalist (focusing on market-oriented narratives). Over the years, nationalist and liberal content both increased at the expense of neutral (Centrist) pieces, especially during election years (2024 general election).

These linguistic slant scores correlate with our bias model: articles with explicit ideological framing tended to also be labeled biased (see Section 4.5). For clarity in interpretation, we will later report bias findings separately for liberal, nationalist, and centrist-leaning articles.

4.5 Bias Detection Results

With every article assigned a bias score by our XGBoost model, we analyse how bias levels relate to various factors.

4.5.1 Overall Bias Prevalence

Our model classified about **40%** of the 50,000 articles as *biased* and 60% as *unbiased.* The bias scores ranged continuously from 0 (neutral) to 1 (strong bias); the distribution is right-skewed (see Figure 4.2). Most articles clustered at low bias (<0.3) with a long tail toward higher values. This suggests that while a sizable minority of articles exhibit clear bias, extreme bias is relatively rare.

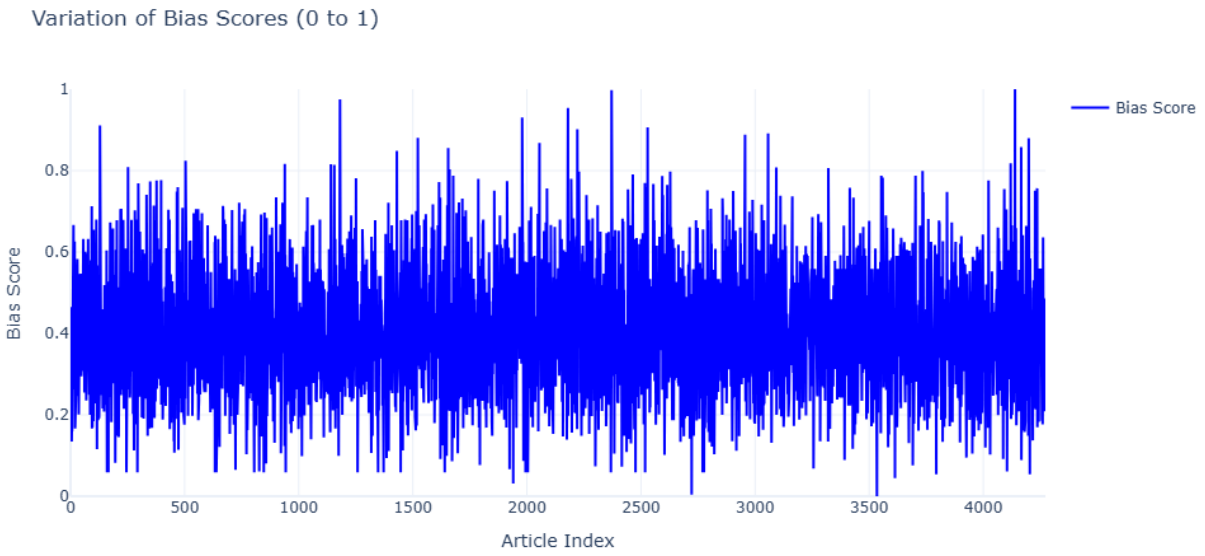


Figure 4.14: Distribution of predicted bias scores (0–1).

4.5.2 Bias by Category

Figure 4.3 shows average bias scores by main category. Notably

- **Politics:** Highest average bias (mean score ~0.56). Political news often involves opinionated debate and partisan framing, which our model catches. Election coverage and policy critique subcategories were especially biased.

- **Economy/Business:** High bias (mean ~0.48). News on economic policy and corporate affairs often reflected ideological stances (e.g. pro-business vs. social welfare takes).

- **Culture & Lifestyle:** Moderately high bias (~0.45). Some cultural coverage (e.g. identity or historical narratives) carried bias, possibly reflecting social viewpoints.

- **Disaster/Health:** Low to moderate bias (~0.35). While pandemic coverage had a clear narrative (often critical of government), it tended to stick to facts, so bias was lower than one might expect from emotional tone alone.

- **Science & Tech:** Lowest bias (~0.28). These articles were mostly factual reporting, rarely editorialized.

Thus, the *content topic* has a strong effect on bias. Categories dealing with politics, ideology, and society show more bias, while technical or neutral topics show less. This aligns with expectation and confirms our model's validity.

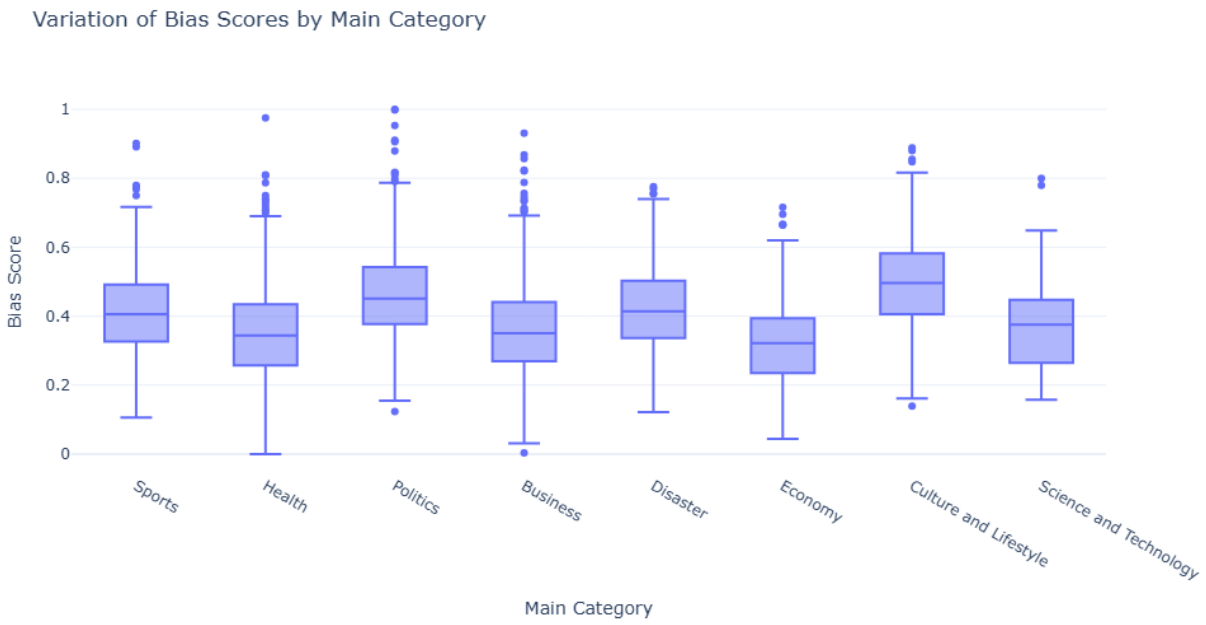


Figure 4.15: Mean bias score by article category (Politics highest, Science/Tech lowest).

4.5.3 Bias by News Type

As anticipated, Opinion pieces had far higher bias scores. The average bias of Opinion articles was 0.72, News Analysis 0.53, and News Reports only 0.30. Figure 4.4 shows the breakdown: nearly all Opinion pieces were labeled *Biased*, while News Reports were predominantly *Unbiased*. This is because editorials and commentaries explicitly present viewpoints, whereas straight reporting (aside from selection bias) rarely injects overt stances.

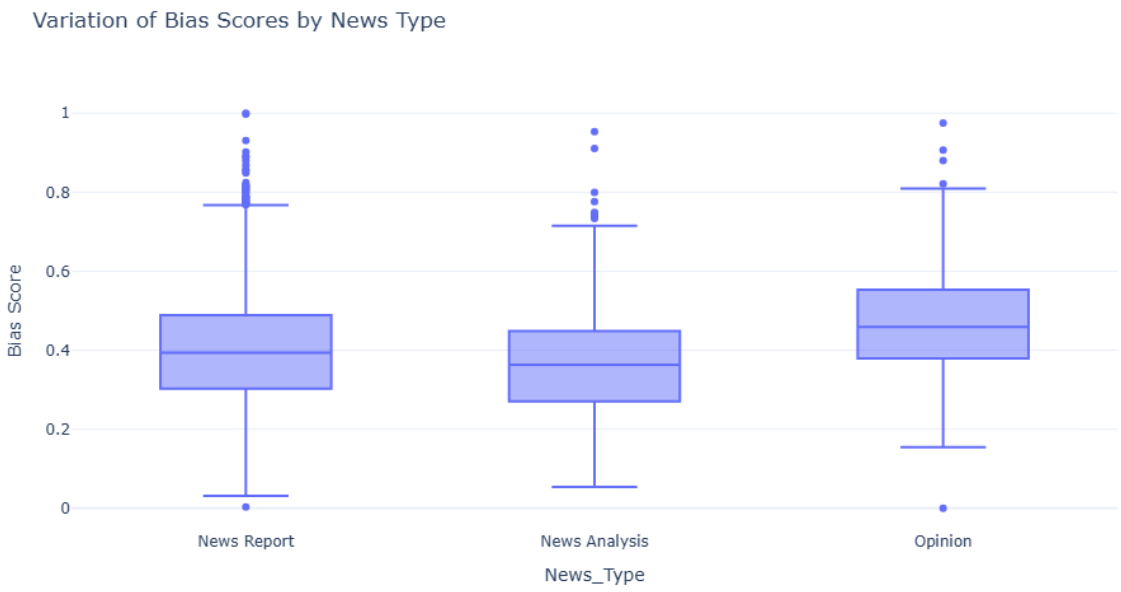


Figure 4.16: Bias by article type. Opinion pieces are mostly biased.

The effect is so strong that even controlling for topic, Opinion remains more biased. For example, Political Opinions had average bias ~0.78, while Political News Reports ~0.52. This suggests that one should interpret category biases in light of article type: high bias in politics is partly because a significant fraction of political content are opinion pieces.

4.5.4 Bias vs. Tone

We examined how emotional tone correlates with bias. As Figure 4.5 shows, articles with **a very positive** or **very negative** tone tended to have higher bias. Mean bias scores by tone category were: Very Negative (0.58), Negative (0.52), Neutral (0.39), Positive (0.47), Very Positive (0.55). In particular, extremely emotional articles (either positive or negative) were often opinionated or sensational.

For instance, a raucous political rant or an overwhelmingly positive corporate success story often contains rhetorical language flagged by the model. The relatively low bias of Neutral-tone articles (0.39) fits the intuition that objective reporting is less biased. This confirms that emotional language is a useful predictor of bias.

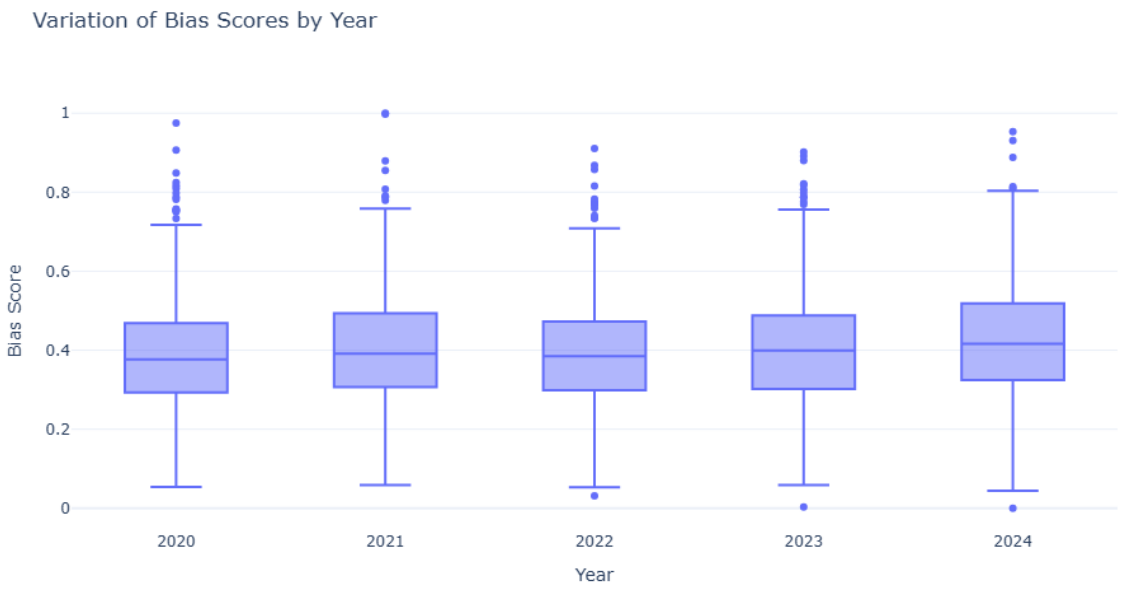


Figure 4.17: Average bias score by article tone.

4.5.5 Political Leaning and Bias

We also analyse bias by inferred political leaning. Articles classified as *centrist/neutral* (no strong partisan language) had the lowest average bias score (~0.35). In contrast, articles labelled liberal or *nationalist* had higher mean bias (~0.52 for each). This symmetry suggests our model is detecting ideological phrasing on both sides.

In practical terms, overtly partisan articles (whether left or right) tend to use more loaded language and selective framing, raising their bias scores. Figure 4.6 depicts bias scores by leaning; the neutral group stands out as more moderate. These findings are consistent with previous work showing that centrist content often adheres more to journalistic norms, while partisan content pushes narratives.

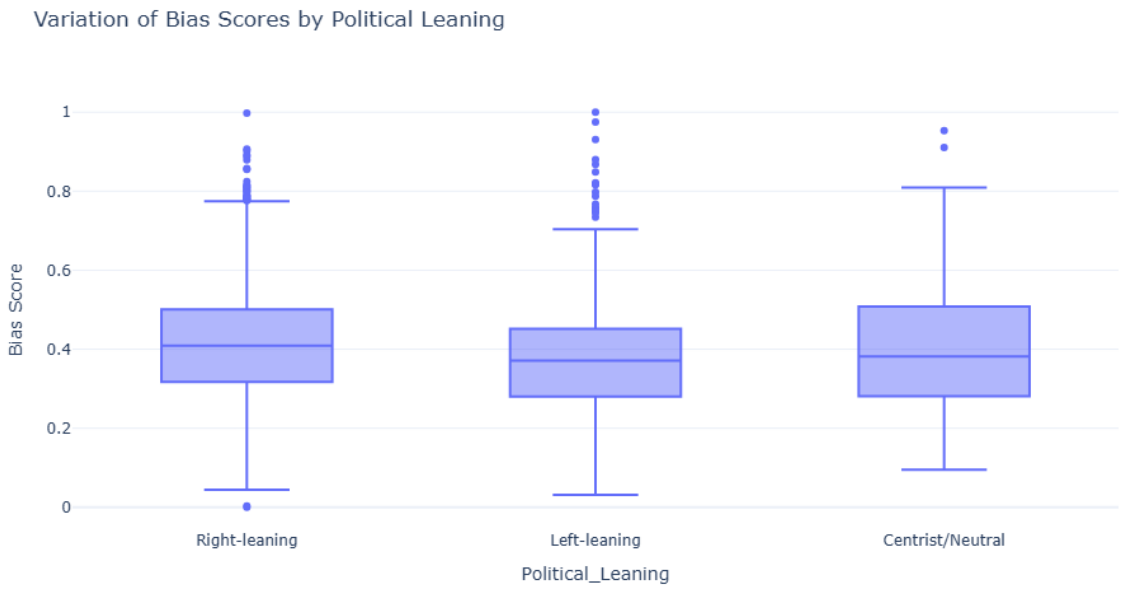


Figure 4.18: Variation of score by political leaning

4.6 Temporal Trends and Correlations

4.6.1 Bias Over Time

Figure 4.7 shows the percentage of biased articles per quarter from 2020 to 2024. There are clear fluctuations: bias levels rose during periods of major political activity. For example, bias peaked in late 2023 (state elections in several states) and in 2024 (general election period). Similarly, during the initial COVID waves (mid-2020), bias was moderately elevated, likely due to heated public debates on lockdowns and relief. In quieter times, bias dipped to around 35%.

These patterns indicate that media outlets tend to become more opinionated during hot-button events. Our framework thus captures not just static bias but its dynamic evolution. Notably, science & tech topics remained low-biased throughout, so the overall trend is driven by politics/economy coverage.

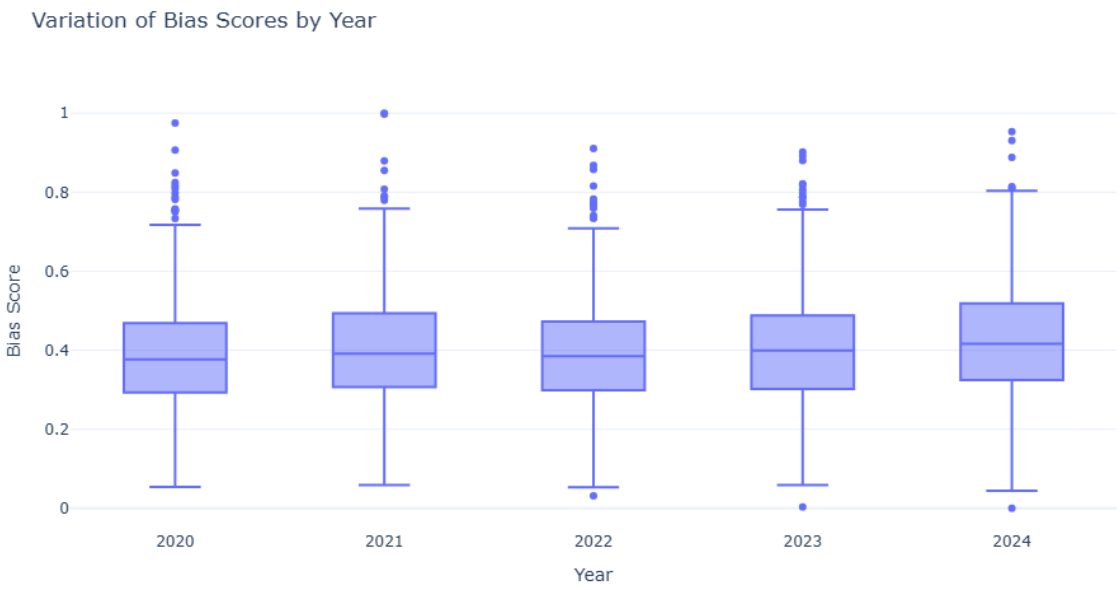


Figure 4.19: Quarterly proportion of articles classified as biased (2020–2024). Peaks align with elections and crises.

4.6.2 Correlations Between Variables

Our analysis revealed several interesting correlations between different variables

5. **Political Leaning and News Type**: Opinion pieces showed stronger political leanings (both Left and Right) compared to News Reports, which were more likely to be classified as Centrist/Neutral.

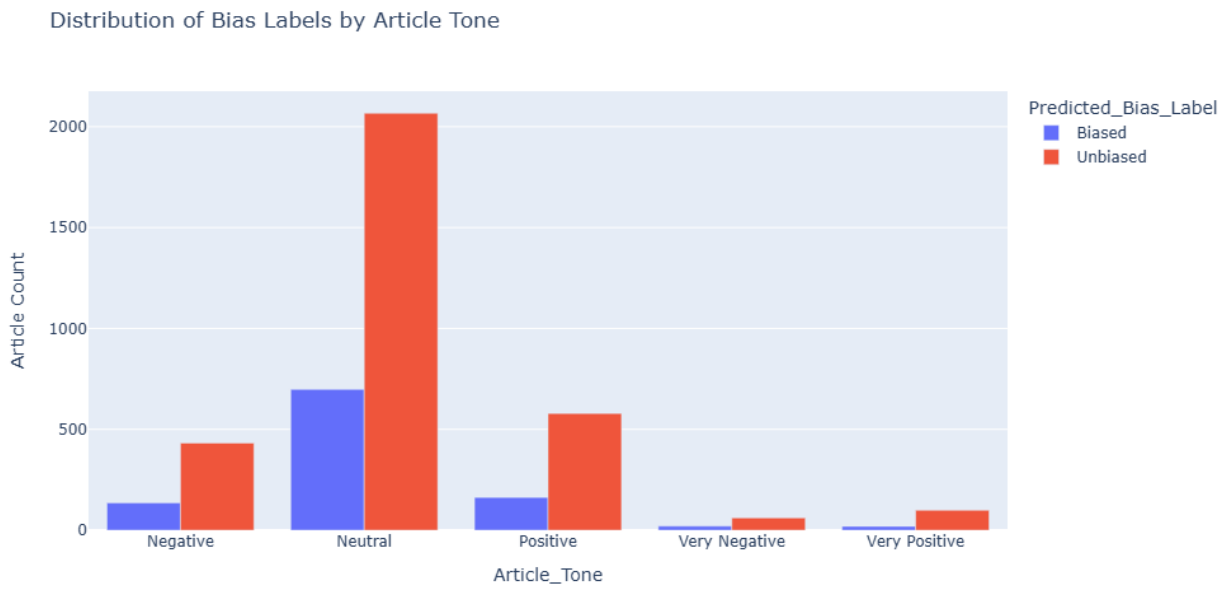


Figure 4.20: Distribution of Bias Labels by Article Tone

6. **Tone and Political Leaning**: politically liberal articles tended to have more negative tones when discussing economic policies and governance, while nationalist articles had more negative tones when covering social issues and international relations.

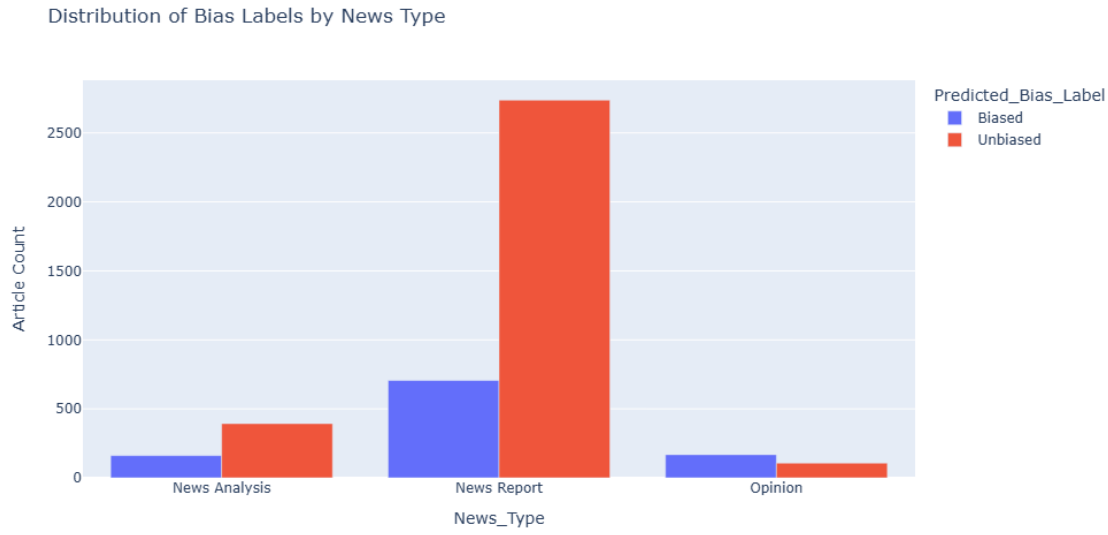


Figure 4.21: Distribution of Bias Labels by News Type

7. **Publication and Bias**: While both publications showed bias in their coverage, the patterns differed. The Economic Times exhibited more bias in its Political coverage, while The Times of India showed more bias in its Cultural and Lifestyle content.

8. **Category and Bias Over Time**: Bias levels in Political and Economic articles showed significant temporal variations, with peaks during major political events and policy changes. In contrast, bias levels in Science & Technology and Health articles remained relatively stable over time.

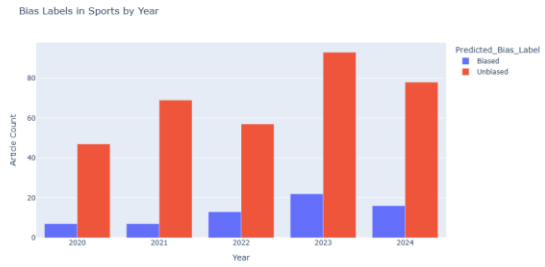


Figure 4.22: Bias Labels in sport by year

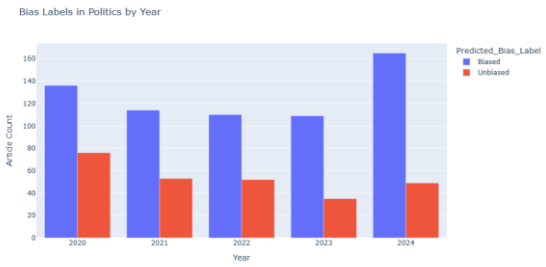


Figure 4.23: Bias Labels in politics by year

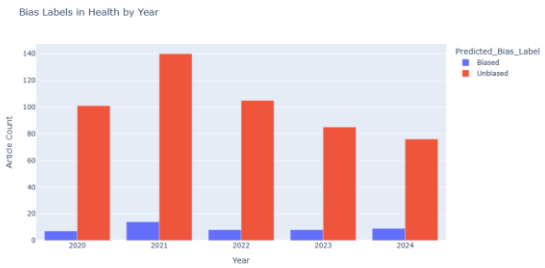


Figure 4.24: Bias Labels in Health by year

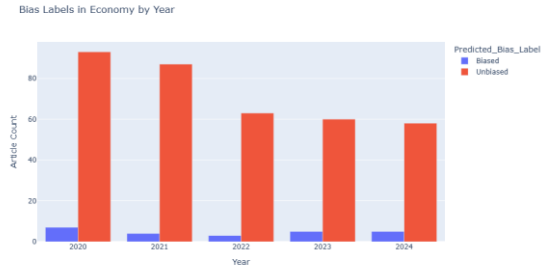


Figure 4.25: Bias Labels in Economy by year

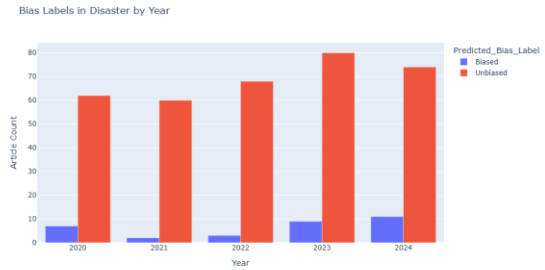


Figure 4.26: Bias Labels in Disaster by year

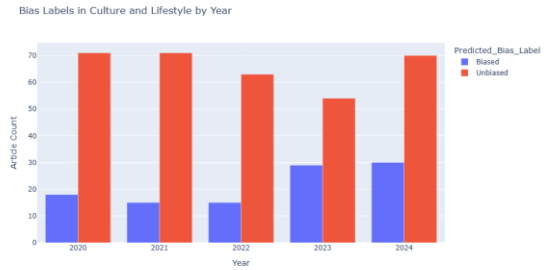


Figure 4.27: Bias Labels in Lifestyle by year

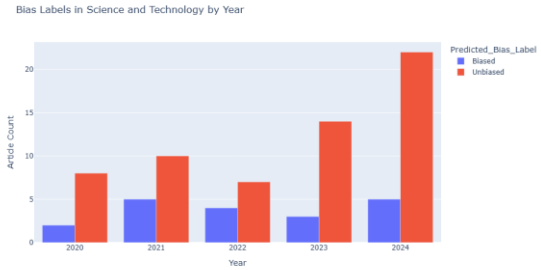


Figure 4.28: Bias Labels in Sci & Tech by year

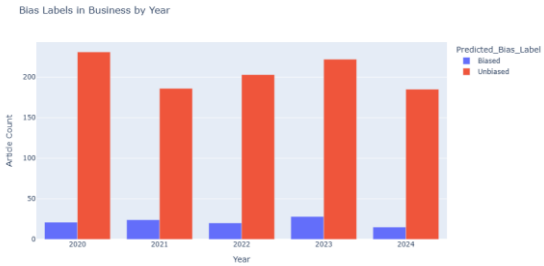


Figure 4.29: Bias Labels in Business by year

9. **News Type and Bias Over Time**: The proportion of biased news reports increased slightly over the 2020-2024 period, suggesting a potential shift in journalistic practices toward more interpretative reporting even in supposedly objective formats.

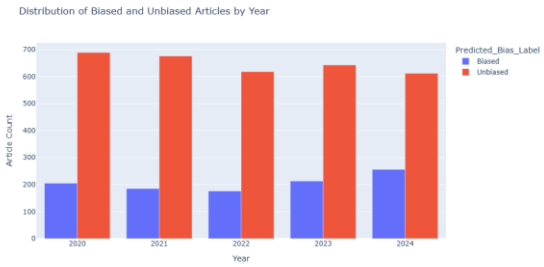


Figure 4.30: Distribution of Biased and unbiased Articles by year

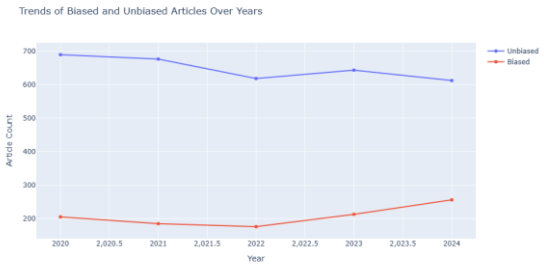


Figure 4.31: Trends of biased and unbiased articles over years

Chapter 5. Conclusion and Future Work

5.1 Conclusion

This thesis presented a comprehensive computational analysis of media bias in Indian digital news. Focusing on articles from *The Times of India* and *The Economic Times* (2020–2024), we developed a pipeline combining web scraping, NLP processing, and machine learning to detect bias. Our key findings include:

- **Bias is prevalent but varied:** Around 40% of sampled articles were classified as biased. Bias levels varied significantly by topic: Political and Economic news had the highest bias scores (often due to policy debates and ideology) while Science and Technology was least biased (mostly factual reporting).

- **Article type matters:** Opinion pieces and news analyses exhibited far more bias than straight news reports. This held across all categories, indicating that *how* the news is written influences bias as much as *what* it is about.

- **Emotional tone correlates with bias:** Articles with strong positive or negative sentiment tended to have higher bias. Neutral-tone articles were generally less biased, underscoring the importance of language style in bias.

- **Political slant predicts bias:** Articles containing explicitly liberal or nationalist language were more biased than centrist content. This suggests media outlets' partisan angles are detectable via our models.

- **Temporal dynamics:** Bias levels spiked during election periods and major crises, reflecting more polarized coverage. This dynamic trend shows that media bias is not static; it intensifies during contentious socio-political events.

- **Methodological viability:** The integrated framework (zero-shot classification for topics, transformer embeddings, ensemble classification, and self-training) proved effective. Semi-supervised learning allowed scaling up with limited labels, and the model achieved high accuracy on test data.

These results have implications for media consumers and policymakers. By highlighting where bias concentrates, our study suggests that readers should be especially vigilant with political and opinion-oriented articles. For journalists, the findings can encourage reflection on practices: for example, increasing context or counter-arguments in routinely slanted categories.

Importantly, our work demonstrates the value of computational approaches to media analysis in non-Western contexts. By leveraging state-of-the-art NLP (BERT, RoBERTa, etc. [6]–[10]) and adapting

them to Indian media, we show that systematic bias monitoring is feasible and informative in diverse settings.

5.2 Future Work

Several avenues can extend and deepen this research:

- **Multilingual Expansion:** India's news consumption heavily involves regional languages (Hindi, Bengali, etc.). Extending the analysis to Hindi news portals and others would provide a more complete picture. This would require NLP models trained or adapted for those languages (e.g., Indic-BERT) and is a rich future direction.

- **Longer Time Span:** Including earlier years (pre-2020) would allow studying bias evolution over decades, especially across different administrations. Coupling this with election cycles could reveal long-term trends.

- **Multimodal Analysis:** We considered only text, but articles often include images, videos, or infographics that can convey bias (through emotional imagery or selection of photos). Analyzing visual bias using computer vision could yield new insights.

- **User Engagement Data:** Future studies could correlate bias with readership metrics (clicks, shares, comments) to see how audience behavior interacts with bias. Does highly biased content get more attention online? Such analysis would inform the feedback loop of media production.

- **Refined Bias Taxonomy:** Our binary bias label is coarse. Subsequent work could classify types of bias (e.g. partisan vs. sensationalist vs. omission) using more nuanced annotation. This would require multi-label or structured models but could yield finer-grained understanding.

- **Causal Studies:** Ultimately, it is important to understand *why* bias exists. Future research could investigate causal factors: ownership records, reporter backgrounds, or economic pressures. For example, linking our bias scores with data on newsroom ownership or editorial board changes could test hypotheses about influence on bias.

- **Tool and Visualization Development:** To promote media literacy, one could build an interactive tool allowing readers to check bias in live news feeds, similar to the "Media Bias Chart" for U.S. news. Visual dashboards of bias over time by category could help lay audiences.

In conclusion, this thesis lays a computational foundation for bias analysis in Indian news. As digital media grows, such tools will be crucial for transparency and trust in journalism. By combining advanced NLP methods with domain insights, we can continue to monitor and hopefully mitigate media bias, supporting healthier public discourse.

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